

# NALLABATI VENKATA DURGA SAI SARAT CHANDRA

Target Role: Robotics Software Engineer - Controls & Middleware • Waymo Autonomy Labs

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## EDUCATION & NATIONAL MERIT PEDIGREE

### Amrita Vishwa Vidyapeetham

August 2024 – Expected May 2028

B.Tech in Robotics & Artificial Intelligence (Cumulative GPA: 9.05 / 10.00, Top 5% Cohort)

Amritapuri, India

- **Academic Standing:** AP POLYCET State Merit Rank: 497 (Top 0.5% Statewide Percentile).
- **National Distinctions:** ISRO-IIRS YUVIKA (Selected for space science analytics) | NAEST National Physics Test (Cleared Prelims & Finals, IAPT / Prof. H.C. Verma) | MTSO Olympiad (Level 2 Qualifier).

## PEER-REVIEWED PUBLICATIONS & PROCEEDINGS

- N. V. D. S. Sarat Chandra, M. Nunna, Asha C. A., P. C. Sreekumar, “Small Scale Raspberry Pi Based Autonomous Car: A Practical Guide to Understand Complexity.” in *Proc. IEEE ICRM*, 2025.
- N. V. D. S. Sarat Chandra, et al., “Physics-Guided Diffusion with Thermodynamic Awareness for 3D Molecular Generation.” Submitted to *IEEE IES*, 2026.
- N. V. D. S. Sarat Chandra, et al., “Domain-Aware Multi-Modal Industrial Diagnostics: Integrating Mechanical Heuristics with Conformal Uncertainty.” Submitted 2026.

## TARGETED ENGINEERING PROJECTS & SYSTEMS ARCHITECTURE

### Industrial GSM & 4G LTE Cellular IoT Gateway with Dual-Mode Failover

Embedded Firmware Engineer

Valetron A7670C 4G LTE Cat-1, ESP32, FreeRTOS, Circular DMA UART, 240V Relays

- **Core Impact:** <1.8s automatic network failover (WiFi <-> 4G LTE); opto-isolated 240V relay switching with RC snubbers.

### Multi-Parameter Health Vitals Monitor & Real-Time IoT Data Streamer

Biomedical Systems Developer

MAX30102 PPG, ESP8266/ESP32, Fast-Mode I2C, 2nd-order IIR Bandpass, FreeRTOS

- **Core Impact:** <200ms real-time SpO2/HRV cloud telemetry; 9-clock I2C bus lockup recovery.

### Pasupatha Autonomous Companion Robot Control Bus

Head of Embedded Systems, Club Mirai

ROS 2 Jazzy, STM32, ESP32, CAN 2.0B, C++ DLS Inverse Kinematics, HIL Testbed

- **Core Impact:** 50 Hz closed-loop control; <50us 5-DOF IK compute cycles (<0.4 deg repeatability).

## TECHNICAL SKILLS & CORE COMPETENCIES

**Languages & Runtimes:** C (C99/C11), C++ (C++17/20), Python (3.12), Bash, MATLAB, POSIX Sockets, FreeRTOS, Bare-Metal

**Robotics, AI & Middleware:** ROS 2 (Jazzy), CycloneDDS, Forward/Inverse Kinematics, OpenCV, PyTorch, PINNs, ONNX Runtime, Vitis AI EP

**Hardware, Buses & Silicon:** CAN Bus (2.0B), UART DMA, I2C (9-clock recovery), SPI, Valetron A7670C (4G LTE), STM32, ESP32, Raspberry Pi 5

**Simulation & Tooling:** ANSYS Fluent (CFD), SolidWorks, Fusion 360, Linux (Ubuntu Server), Docker, Git/GitHub, LaTeX